We have already found power series for the radial part of the wavefunction assuming a separable solution and including only quantum number n: ignoring dependence on angular momentum l and magnetic quantum number m. The angular wavefunction turns out to be related to the spherical harmonics, a set of special functions and the power series we have already found turns out to be related to the 'laguerre polynomials.' Since these mathematical objects are unfamiliar we will not derive the full wavefunction but will show how we can include dependence on angular momentum to the radial wavefunction using the same power series method as before.

1 Angular momentum

To include dependence of angular momentum is a good excuse to review it in quantum mechanics. In classical mechanics angular momentum is a vector defined by $\vec{L} = \vec{r} \times \vec{p}$. To convert this to quantum mechanics we promote measurable quantities to operators: $\hat{p} = -i\hbar\frac{d}{dx}$ and $[\hat{x}, \hat{p}] = \hat{x}\hat{p} - \hat{p}\hat{x} = i\hbar$. Our aim is to relate the components of angular momentum in the cartesian directions: $\hat{L}_x, \hat{L}_y, \hat{L}_z$ to the magnitude $\hat{L}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2$.

As usual with quantum mechanics we can learn alot from commutation relations:

$$[L_x, L_y] = [yp_z - zp_y, zp_x - xp_z]$$
$$= [yp_z, zp_x] + [zp_y, xp_z] - [zp_y, zp_x] - [yp_z, xp_z]$$

Note that things commute with themselves so:

$$[L_x, L_y] = [yp_z, zp_x] + [zp_y, xp_z]$$
$$= yp_z zp_x - zp_x yp_z + zp_y xp_z - xp_z zp_y$$
$$= i\hbar(yp_x - p_x y + p_y x - xp_y)$$
$$= i\hbar\hat{L}_z$$

Since the components of angular momentum do not commute we cannot find wavefunctions that are eigenstates of both. It turns out that the magnitude of angular momentum, L does commute with the components:

$$[L, L_x] = [L_x^2, L_x] + [L_y^2, L_x] + [L_z^2, L_x]$$

$$= L_y(-i\hbar L_z) + (-i\hbar L_z)L_y + L_z(i\hbar L_y) + (i\hbar L_y)L_z$$

$= 0$

This means we can look for eigen states of one component of angular momentum, say L_z, and the magnitude L. Lucky for us, there are creation and annihilation operators for angular momentum:

$$L_+ = L_x + iL_y$$
$$L_- = L_x - iL_y$$

$$[L_z, L_+] = [L_z, L_x] + i[L_z, L_y] = i\hbar L_y - i^2\hbar L_x = \hbar L_+$$

Now let M be an eigenfunction of L_z and L with z component of angular momentum $m\hbar$. Consider L_+M.

$$L_z(L_+M) = ([L_z, L_+] + L_+L_z)M = (\hbar L_+ + m\hbar L_+)M$$
$$= \hbar(m+1)L_+M$$

It is another eigenfunction with eigenvalue $\hbar(m+1)$. Thus we can interpret L_+ as adding a quanta of angular momentum. Similar calculations can be done to show L_- removes a quanta of angular momentum.

2 Separation of variables

In spherical co-ordinates, the schrodinger equation reads:

$$-\frac{\hbar^2}{2m}[\frac{1}{r}\partial_r^2(r\psi) + \frac{1}{r^2\sin^2\theta}\partial_\theta(\sin\theta\partial_\theta\psi) + \frac{1}{r^2\sin^2\theta}\partial_\phi^2\psi] - \frac{e^2}{4\pi\varepsilon_o}\psi = E\psi$$

We want to look for a separable solution of the form :

$$\psi_{n,l,m} = aY_{l,m}(\theta,\phi)F_l(r)$$

where a is a normalisation constant that depends on the definition chosen for $Y_{l,m}$. As said at the start, we aim to merely find $F_l(r)$ not $Y_{l,m}(\theta,\phi)$.

3 Angular momentum

In classical mechanics angular momentum is a vector defined by $\vec{L} = \vec{r} \times \vec{p}$. If we promote measurable quantities to operators: $\hat{p} = -i\hbar\frac{d}{dx}$ and $[\hat{x}, \hat{p}] = \hat{x}\hat{p} - \hat{p}\hat{x} = i\hbar$. Our aim is to relate the components of angular momentum in the cartesian directions: $\hat{L_x}, \hat{L_y}, \hat{L_z}$ to the magnitude $\hat{L}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2$.

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Now let M be an eigenfunction of L_z and L with z component of angular momentum $m\hbar$. Consider L_+M.

$$\begin{aligned} L_z(L_+M) &= ([L_z, L_+] + L_+L_z)M = (\hbar L_+ + m\hbar L_+)M \\ &= \hbar(m+1)L_+M \end{aligned}$$

It is another eigenfunction with eigenvalue $\hbar(m+1)$. Thus we can interpret L_+ as adding a quanta of angular momentum. Similar calculations can be done to show L_- removes a quanta of angular momentum.

4 Including angular momentum

In spherical co-ordinates, the schrodinger equation reads:

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We want to look for a separable solution of the form :

$$\psi_{n,l,m} = aY_{l,m}(\theta,\phi)F_l(r)$$

where a is a normalisation constant that depends on the definition chosen for $Y_{l,m}$. We aim to find $F_l(r)$.

Instead of plugging in this separable solution into the schrodinger equation we will take a shortcut to find a differential equation for $F_l(r)$.

In a classical central forces problem angular momentum is conserved because:

$$\begin{aligned} \frac{d\vec{L}}{dt} &= \frac{d}{dt}(\vec{r}\times\vec{p}) \\ &= \frac{d\vec{r}}{dt}\times\vec{p} + \vec{r}\times\vec{F} \end{aligned}$$

but $\vec{p} = m\vec{v}$ so momentum and velocity are parallel making the first term vanish. If the force is central it only depends on radial distance so the second term also vanishes since force and displacement are also parallel.

If we let the orbit occur in the x-z plane then the energy is:

$$E = T + V = \frac{1}{2}m\dot{r}^2 + \frac{1}{2}mr^2\dot{\theta}^2 + V$$

but in this case $L = mr^2\dot{\theta}$ and so

$$E = \frac{1}{2}m\dot{r}^2 + \frac{L^2}{2mr^2} + V$$

which looks like a 1D problem with the "effective" potential $\frac{L^2}{2mr^2} + V$

Applying this potential to the schrodinger equation we promote angular momentum to an operator, $\hat{L}$ and us the result we already derived for $\hat{L^2}F = \hbar^2 l(l+1)F$. (Recall F is the radial part of our separable solution.)

$$-\frac{\hbar^2}{2m}\frac{1}{r}\frac{d^2(rF)}{dr^2} - \frac{e^2}{4\pi\varepsilon_0 r^2}F + \frac{\hbar^2 l(l+1)}{2mr^2}F = EF$$

Using the convenient bohr radius/energy as earlier and subbing $f = rF$:

$$f'' = -(\varepsilon + \frac{2}{\rho} + \frac{l(l+1)}{\rho})f$$

which means when we replace f with g and expand in a power series in ρ there is an extra $-l(l+1)$ in the ρ^{-1} coefficient.

Demanding all the terms vanish so all the coefficients vanish:

$$k(k+1) - l(l+1)a_{k+1} - 2(\alpha k - 1)a_k = 0$$

recall $\alpha^2 = -\varepsilon$

We have a slightly modified recursion relation:

$$a_{k+1} = \frac{2(\alpha k - 1)}{k(k+1) - l(l+1)}\alpha_k$$

note there is a new restriction since if we demand the denominator is non-zero $k > l$ so energy levels start at $n = l + 1$, which we can interpret as having enough energy to have that angular momentum.

5 The ground state of Helium

Unfortunately, there no closed form solution has been found to the Schrodinger equation for helium. We can still use the solution for hydrogen to approximate solutions for larger elements. One approximation method is called the variational principle and states that:

$$\langle\psi|H|\psi\rangle > E_0$$

where E_0 is the ground state energy and ψ is any wavefunction. This means that using wavefunctions that have nothing to do with the system being studied excellent approximations to the ground state energy can be made by trying to lower the upper bound $\langle\psi|H|\psi\rangle$. This is best used as a method for approximating energies, not wavefunctions, but we will use it here since it is a relatively simple way of modelling helium The full hamiltonian for helium (an atom with two protons, some neutrons and two electrons) is:

$$H = \frac{\hbar^2}{2m}(\nabla_1^2 + \nabla_2^2) - \frac{2ke^2}{\vec{r_1}} - \frac{2ke^2}{\vec{r_2}} - \frac{ke^2}{|\vec{r_2} - \vec{r_1}|}$$

In english: the electrons are located at $\vec{r_1}$ and $\vec{r_2}$ with the nucleus the origin. There are two kinetic terms for the electrons, two terms for the electromagnetic electron-nucleus interaction and one for the electron electron interaction. Here m really should be a reduced mass m $= \frac{Am_em_p}{m_e+Am_p}$ where A is the atomic number and will vary depending on the number of neutrons in the helium nucleus.

The hamiltonian naturally resembles the one for hydrogen we saw at the start, only the electron-electron interaction is new. If we ignore it altogether we have two independent hydrogen hamiltonians and so the helium ground-state wavefunction will be a product of two hydrogen ground-state wavefunctions:

$$\psi_0 = (\frac{1}{\sqrt{\pi a^3}})^2 e^{-r_1/a} e^{-r_2/a} = \frac{1}{\pi a^3} e^{-r_1/a - r_2/a}$$

if we take a_B to be a bohr radius then we have $a = a_B/2$ since a has an e^2 factor in the denominator that doubles since there are two protons. Since ψ_0 is an eigenfunction of the non interacting hamiltonian:

$$H\psi_0 = (2^3 E_R + V_{ee})$$

where $E_R \approx 13.6eV$is the Rydberg constant and is minus the ground state energy for hydrogen. ψ_0 has a ground state energy $E \approx -13.6 \times 8 = -109eV$. Here we will make this energy slightly more accurate to the actual ground state ($\approx -79eV$) by varying the charge in the wavefunction Z and finding what Z minimises $\langle\psi|H|\psi\rangle$. This is a crude model of charge screening: one electron experiences a lower "effective' charge at the nucleus due to repulsion from the other.

A suitable wavefunction is:

$$\psi = \frac{Z^3}{\pi a_B^3} e^{-Zr_1/a_B - Zr_2/a_B}$$

Since, by scaling, this is an eigenfunction of the hamiltonian:

$$H' = \frac{\hbar^2}{2m}(\nabla_1^2 + \nabla_2^2) - \frac{kZe^2}{\vec{r_1}} - \frac{kZe^2}{\vec{r_2}}$$
$$H'\psi = Z^2 E_R \psi$$

and so to use this result to evaluate $\langle\psi|H|\psi\rangle$, it will be convenient to rewrite the actual hamiltonian for heium H as

$$H = H' + \frac{ke^2}{|\vec{r_2} - \vec{r_1}|} + \frac{k(2-Z)e^2}{\vec{r_1}} - \frac{k(2-Z)e^2}{\vec{r_2}}$$

so

$$\langle\psi|H|\psi\rangle = Z^2 E_R + ke^2 \langle\psi|\frac{1}{|\vec{r_2} - \vec{r_1}|}|\psi\rangle - (2-Z)ke^2\langle\psi|\frac{1}{r}|\psi\rangle$$